

IEOR 4004 Project 1: Child Care Deserts

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1 Overview

In this work, we present our approach and solutions for *Project I: Eliminating Child Care Deserts in New York State through Optimization*. In the first section, we provide a brief description of what constitutes a child care desert. We then outline our assumptions regarding data processing and cleaning before presenting the formulations and solutions to **The Problem of Budgeting** and **The Problem of Realistic Capacity Expansion and Location**. Finally, we provide meaningful insights on the results of our models.

2 Introduction

As explained in the project description, the term *child care desert* depends on the total number of available child care slots, the population of children requiring care, the percentage of employed parents, and the average income in the area.

We then classify an area as a desert based on its demand for child care:

1. **High-demand areas:** Demand is considered high in regions, specifically zip codes, where at least 60% of parents are employed or the average household income is \$60,000 or less per year. These areas are considered to be a child care desert if:

$$\text{Number of available slots} \leq \frac{\text{Population of kids aged two weeks to 12 years}}{2} \quad (1)$$

2. **Normal-demand areas:** Demand is considered normal in zip codes where employment and income levels do not meet the high-demand criteria. These regions are defined as a child care desert if:

$$\text{Number of available slots} \leq \frac{\text{Population of kids aged two weeks to 12 years}}{3} \quad (2)$$

Moreover, the New York State (NYS) government requires that **all** children under the age of 5 have sufficient access to care. Therefore, the number of available slots for children in this age group must be defined as follows:

$$\text{Number of available slots}_{0-5 \text{ years}} < \frac{2}{3} \cdot \text{Population}_{0-5 \text{ years}} \quad (3)$$

To eliminate child care deserts in New York, we build an integer programming model to determine the optimal number of new facilities to build and the expansion levels for existing facilities within each zip code all while minimizing total funding. To accomplish this, the data must be cleaned and processed in order to identify all of the zip codes that are child care deserts. In the following section, we describe our data processing steps and outline the assumptions made during this stage.

3 Data Cleaning

Conducting a background search on how zip codes work in the state of New York, we discovered that certain landmarks and office buildings in Manhattan have their own zip codes because they receive a high volume of mail. Additionally, there are zip codes designed specifically for mail services and are for non-residential use (Shah, 2018). Using this information, we proceeded to remove any zip codes in our datasets that fell under these two categories.¹

While analyzing the data, we also realized that several zip codes were missing employment rate and income data. To address these missing values, we imputed data at the zip code prefix level by grouping zip codes by their first three digits and calculating the average employment rate and average income for each group. If a zip code was missing either employment rate or income, we replaced its missing value with the corresponding

¹Refer to the file `Commercial zip codes.csv` for the list of these zip codes. In our data cleaning Jupyter notebook, we removed all zip codes listed in this file.

group average. For example, if the zip code 10001 had missing data, its corresponding employment rate and income values would be the mean employment rate and mean income of all zip codes starting with 100.

For the population dataset, we needed to estimate the total number of children aged two weeks to 12 years within each zip code. However, the dataset provided only reports population counts in the following age groups: 0–5, 5–9, 10–14, and so forth. We assumed that the number of infants aged 0 to 2 weeks old within the 0–5 age group would be negligible and therefore did not make any adjustments to this category. To approximate the number of 10 to 12 year olds, we made an assumption based on the proportion that 3 out of the 5 years in the 10–14 age group would correspond to ages 10 to 12.

In summary, the calculation for the total number of children was:

$$\text{Children}_{2w-12} = \text{Population}_{0-5} + \text{Population}_{5-9} + 0.6 \times \text{Population}_{10-14} \quad (4)$$

To address the missing population-by-age-group data for certain zip codes, we sourced a 2023 dataset from the U.S. Census Bureau that reports population counts by age group for zip codes in New York. While this data does not account for all of the zip codes we had missing data for, it did provide data for 151 out of the 458 affected zip codes.² After replacing the missing population data with the values from this public dataset, we decided to delete the remaining 307 zip codes. We chose not to apply the same data imputation method used for handling employment rate and income data because the variation across zip codes could have been higher when looking at population by age group.

For the child care center dataset, we observed that several existing facilities were missing longitude and latitude coordinates. Because each facility belongs to a unique geographic location, replacing the missing coordinates with a summary statistic would not lead to meaningful results. Therefore, we removed all zip codes that contained facilities with missing geographic data. Even though Part 1 did not require the use of the longitude/latitude data, we chose to remove these zip codes starting in Part 1 to ensure that both of our models in Part 1 and Part 2 used the same set of zip codes. This allowed for a more accurate and consistent comparison of results.

During our exploratory data analysis, we learned that a large majority of zip codes in the state of New York are classified as child care deserts. From our data cleaning methods, we ended up with a set of 1,470 zip codes. Out of the 1,470, 1,331 zip codes are classified as deserts which make up 89.3% of all zip codes as shown in Figure 1 below. These 1,331 zip codes were the childcare deserts that we aimed to eliminate in our models in parts 1 and 2.

²Refer to the file `Public Data - Population By Age Group.xlsx` for this data (Sheet Name = “Population by Age Group”).

Distribution of Child Care Deserts in New York State

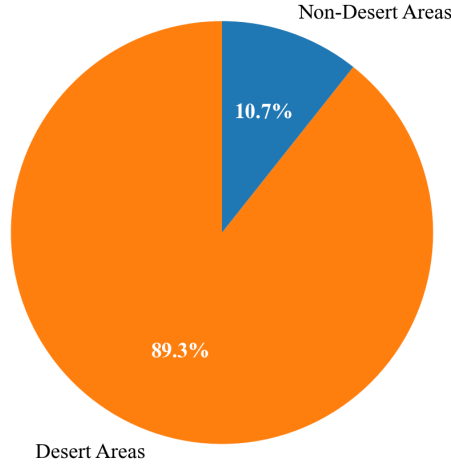


Figure 1: Desert Distribution Across NYS Zipcodes

In addition, to understand what the different capacity columns represented in the childcare center dataset, we found a similar dataset with the same set of columns from the Office of Information Technology Services of the state of New York (New York State Office of Children and Family Services, 2020). The dataset can also be found in the file `NYC Public Database - Child_Care_Regulated_Programs_20251005.csv`. In this dataset, they defined the various columns of capacity as follows:

- **Infant_capacity:** total number of children up to 18 months of age (1.5 years old)
- **Toddler_capacity:** 18 months to 36 months
- **Preschool_capacity:** at least 3 years of age not yet enrolled in kindergarten or a higher age
- **School_age_capacity:** under 13 years of age who are enrolled in kindergarten or a higher age
- **Children_capacity:** ages 6 weeks to 12 years (from the “capacity description”)

We then used this information to approximate the total available capacity of children under 5 years old at each existing facility as:

$$\text{Capacity}_{<5} = \text{Infant_capacity} + \text{Toddler_capacity} + 0.6 \times \text{Preschool_capacity} \quad (5)$$

We also made the assumption that children who make up the preschool capacity would range from ages 3 to 7 to account for children with any special circumstances that may have delayed preschool enrollment. Therefore, 60% of children ages 3 to 7 proportionally corresponds to children aged 3 to 5.

The name of our data cleaning jupyter notebook is `Data Cleaning Part 1.ipynb`, and the output file that corresponds to our data cleaning process is: `Childcare Deserts FINAL.csv`.

4 The Problem of Budgeting

We are tasked with finding the **minimum total funding** required to eliminate *child care deserts* in all zip codes in the state of New York. To do so, we must increase the number of child care slots in each zip code via two options: build new facilities or expand existing facilities.

These expansion and construction options are defined as follows:

- **Build New Facilities**

- **Small:** 100 total slots (up to 50 slots for ages 0–5) for \$65,000.
- **Medium:** 200 total slots (up to 100 slots for ages 0–5) for \$95,000.
- **Large:** 400 total slots (up to 200 slots for ages 0–5) for \$115,000.

- **Expand Existing Facilities**

- Expand by up to 120% of current capacity or up to a maximum of 500 additional slots, whichever is lower.
- If we want to expand capacity by 100% or more, the state must pay a baseline cost of \$20,000, plus \$200 per existing slot.
- New slots for children under five years of age require \$100 per slot for specialized equipment.

These two options for increasing capacity are represented as decision variables in the model, as described in the following section.

4.1 Decision Variables and Assumptions

We define our decision variables to control the number of new facilities to be built and the number of slots that will be added through expansion to existing facilities.

4.1.1 Building

Let us define $B_{s,z}$, $B_{m,z}$, and $B_{l,z}$ as the number of small, medium, and large facilities to be built in zip code z . Then, the cost to build facilities in zip code z is:

$$\text{Cost}_{B,z} = 65,000 \cdot B_{s,z} + 95,000 \cdot B_{m,z} + 115,000 \cdot B_{l,z} \quad (6)$$

where $B_{s,z}, B_{m,z}, B_{l,z} \geq 0$ and must be integers.

Given the following information, we can determine how many of these new slots will be for children aged 0–5.

- A small facility adds 100 slots, with up to 50 slots for ages 0–5.
- A medium facility adds 200 slots, with up to 100 slots for ages 0–5.
- A large facility adds 400 slots, with up to 200 slots for ages 0–5.

For each zip code z , let us define U_z that counts the number of slots specifically for children under five that will be added based on the number of new small, medium, and large buildings to construct in z . Therefore,

$$0 \leq U_z \leq 50B_{s,z} + 100B_{m,z} + 200B_{l,z} \quad (7)$$

4.1.2 Expanding

With regards to choosing which facilities to expand, we need to make assumptions regarding the cost of expansion and its associated constraints.

First, the project description does not specify the cost of expanding a single slot in each facility. We know that if we want to expand capacity by 100% or more, the state must pay a baseline cost of \$20,000 plus \$200 per existing slot. However, there is no clear information on the cost of expanding one slot if we expand by less than 100%.

Therefore, we will use the following formula and variables to model the cost of expansion. Let $x_{f,z}$ be the number of additional slots to be created in the facility f in zip code z , and let $n_{f,z}$ be the current capacity of facility f in zip code z . Then, the cost of expanding $x_{f,z}$ slots in this facility is defined by:

$$\text{Cost}_{f,z} = (20,000 + 200 \cdot n_{f,z}) \cdot \frac{x_{f,z}}{n_{f,z}} \quad (8)$$

where

$$\frac{x_{f,z}}{n_{f,z}} \in \left[0, \min \left\{1.2, \frac{500}{n_{f,z}}\right\}\right].$$

With this formula, the cost of expanding a single slot is well-defined, and the baseline cost of \$20,000 is proportionally allocated across all additional slots. We can then define the total cost of expansion for each zip code z as:

$$\text{Cost}_{E,z} = \sum_{f \in F_z} (20,000 + 200 \cdot n_{f,z}) \cdot \frac{x_{f,z}}{n_{f,z}} \quad (9)$$

where F_z denotes the set of all facilities that can be expanded in zip code z .

Similar to the approach we used for constructing the “building” decision variables, we also need to determine how many of the new expansion slots will be allocated to children under five years old. To capture this, we introduce the decision variable $E_{f,z}$ for each facility f in zip code z , representing the number of additional slots designated for children under five. This decision variable satisfies the following constraint:

$$0 \leq E_{f,z} \leq x_{f,z}$$

Finally, we know that the slots allocated to children under age five have an additional cost of \$100, so the total cost for expansion in zip code z would be:

$$\text{Cost}_{5,z} = 100 \cdot U_z + \sum_{f \in F_z} 100 \cdot E_{f,z}. \quad (10)$$

4.2 Model Formulation

Let us define Z as the set of all the zip codes with child care deserts in the state of New York, $Z_h \subseteq Z$ as the set of zip codes in high demand areas, and $Z_n \subseteq Z$ as the set of zip codes in normal demand areas. Moreover, let us define the following parameters and decision variables.

Parameters

- $n_{f,z}$: current capacity of facility f in zip code z
- F_z : set of existing facilities in zip code z
- P_z : population count of kids aged two weeks to 12 years in zip code z
- $P_{5,z}$: population count of kids aged 0-5 years in zip code z
- ExistingSlots_z : current total slots across all existing facilities in zip code z
- $\text{ExistingSlots}_{5,z}$: current slots across all existing facilities for children aged 0-5 in zip code z

Decision Variables

- $B_{s,z}, B_{m,z}, B_{l,z}$: number of small, medium, and large facilities to build in zip code z
- $x_{f,z} \geq 0$: number of additional slots created by expanding facility $f \in F_z$
- $U_z \geq 0$: number of new slots for children under 5 from newly built facilities in zip code z
- $E_{f,z} \geq 0$: number of new slots for children under 5 from expansion of facility f in zip code z

Objective Function

With these decision variables and parameters, we define the total number of new slots in zip code z as:

$$\text{NewSlots}_z = 100B_{s,z} + 200B_{m,z} + 400B_{l,z} + \sum_{f \in F} x_{f,z}$$

Finally, we define $\text{Cost}_{B,z}$, $\text{Cost}_{E,z}$, and $\text{Cost}_{5,z}$ as the costs associated with building, expanding, and providing slots for children under age five in each zip code z , respectively, as follows:

$$\begin{aligned} \text{Cost}_{B,z} &= 65,000B_{s,z} + 95,000B_{m,z} + 115,000B_{l,z} \\ \text{Cost}_{E,z} &= \sum_{f \in F_z} (20,000 + 200n_{f,z}) \frac{x_{f,z}}{n_{f,z}} \\ \text{Cost}_{5,z} &= 100 \left(U_z + \sum_{f \in F_z} E_{f,z} \right) \end{aligned}$$

Constraints

In addition to the constraints we defined in section 4.1.1 and 4.1.2 for building and expanding, we require the following additional constraints.

For high demand areas to no longer be considered a childcare desert, the number of available slots must be greater than or equal to one-half of the population of children aged 2 weeks to 12 years.

$$\text{ExistingSlots}_z + \text{NewSlots}_z \geq \frac{P_z}{2} \quad \forall z \in Z_h, \quad (11)$$

For normal demand areas to no longer be considered a childcare desert, the number of available slots must be greater than or equal to one-third of the population of children aged 2 weeks to 12 years.

$$\text{ExistingSlots}_z + \text{NewSlots}_z \geq \frac{P_z}{3} \quad \forall z \in Z_n, \quad (12)$$

The number of available slots for children aged 0-5 must also be at least two-thirds of the population of children aged 0-5.

$$\text{ExistingSlots}_{5,z} + U_z + \sum_{f \in F} E_{f,z} \geq \frac{2}{3}P_{5,z} \quad \forall z \in Z. \quad (13)$$

Combining our parameters, decision variables, objective function, and constraints, our optimization model is formulated to be:

$$\begin{aligned} \min \quad & \sum_{z \in Z} (\text{Cost}_{B,z} + \text{Cost}_{E,z} + \text{Cost}_{5,z}) \\ \text{s.t.} \quad & 0 \leq U_z \leq 50B_{s,z} + 100B_{m,z} + 200B_{l,z} \quad \forall z \in Z, \\ & 0 \leq E_{f,z} \leq x_{f,z} \quad \forall f \in F_z \text{ and } \forall z \in Z, \\ & 0 \leq x_{f,z} \leq \min(1.2n_{f,z}, 500) \quad \forall f \in F_z \text{ and } \forall z \in Z, \\ & \text{ExistingSlots}_z + \text{NewSlots}_z \geq \frac{P_z}{2} \quad \forall z \in Z_h, \\ & \text{ExistingSlots}_z + \text{NewSlots}_z \geq \frac{P_z}{3} \quad \forall z \in Z_n, \\ & \text{ExistingSlots}_{5,z} + U_z + \sum_{f \in F} E_{f,z} \geq \frac{2}{3}P_{5,z} \quad \forall z \in Z, \\ & B_{s,z}, B_{m,z}, B_{l,z}, E_{f,z}, x_{f,z}, U_z \in \mathbb{Z}^+ \end{aligned} \quad (14)$$

4.3 Results

After running the optimization model for Problem 1, we determine the minimum total funding required to eliminate all 1,331 identified child care deserts to be **\$222,168,837.22**.

With a 100% desert elimination rate, this strategy relies on a combination of constructing new facilities and expanding existing ones. Construction costs make up 58.1% of the total funding while expansion costs and equipment investments account for 27.2% and 14.7% respectively. The model prioritizes building 706 new large facilities compared to the 160 medium and 503 small facilities. Constructing these large facilities therefore contribute to the majority of the added capacity across the zip codes.

Please refer to `OptimizationBudgeting_Problem1.ipynb` for the code and `optimization_result_problem1_final.csv` for the output file of our model.

Table 1: Results Summary for Problem 1

Metric	Value
Zip Codes Covered	1,313
Desert Elimination Rate	100.0%
Small Facilities (100 slots)	503
Medium Facilities (200 slots)	160
Large Facilities (400 slots)	706
Total Slots from Construction	364,700
Added Under Age 5 Capacity	326,153 slots
Total Existing Capacity	269,790 slots
Total Expanded Capacity	174,507 slots

Table 2: Cost Analysis for Problem 1

Cost Category	Amount	Percentage
Construction Cost	\$129,085,000	58.1%
Expansion Cost	\$60,468,537.22	27.2%
Equipment Cost	\$32,615,300	14.7%
Total Cost	\$222,168,837.22	100.0%

5 The Problem of Realistic Capacity Expansion and Location

NYS officials provided a more detailed and realistic expansion cost function, as well as additional constraints related to the construction of new facilities.

5.1 Decision Variables and Assumptions

In this section, we analyze two new modifications to our model: a new expansion cost function and an additional location constraint.

5.1.1 Cost Function

The officials informed that due to space limitations, it is not possible to expand the capacity of existing facilities by more than 20%, and the marginal cost of expanding slots in existing facilities should vary with the scale of the expansion. Therefore, the cost of expanding the facility f in zip code z shall be defined as:

$$c_{x,f} = \begin{cases} (20,000 + 200n_{f,x}) \frac{x_{f,z}}{n_{f,x}}, & 0 < \frac{x_{f,z}}{n_{f,x}} \leq 0.1 \\ (20,000 + 400n_{f,x}) \frac{x_{f,z}}{n_{f,x}}, & 0.1 < \frac{x_{f,z}}{n_{f,x}} \leq 0.15 \\ (20,000 + 1,000n_{f,x}) \frac{x_{f,z}}{n_{f,x}}, & 0.15 < \frac{x_{f,z}}{n_{f,x}} \leq 0.2 \end{cases} \quad (15)$$

This is a nonlinear piecewise function, and therefore, to retain our model as a mixed integer program, we need to introduce new decision variables. Let us focus on the cost of expansion of only one facility before generalizing for all facilities.

Let us define $\delta_1, \delta_2, \delta_3$ such that:

$$\frac{x_{f,z}}{n_{f,z}} = \delta_1 + \delta_2 + \delta_3$$

and

$$0 \leq \delta_1 \leq 0.1, \quad 0 \leq \delta_2 \leq 0.05, \quad 0 \leq \delta_3 \leq 0.05$$

Then, the total expansion cost can be expressed as:

$$c_{f,z} = (20,000 + 200n_{f,z})\delta_1 n_{f,z} + (20,000 + 400n_{f,z})\delta_2 n_{f,z} + (20,000 + 1,000n_{f,z})\delta_3 n_{f,z} \quad (16)$$

We then introduce two binary variables $w_1, w_2 \in \{0, 1\}$ such that:

$$\begin{aligned} 0.1w_1 &\leq \delta_1 \leq 0.1, \\ 0.05w_2 &\leq \delta_2 \leq 0.05w_1, \\ 0 &\leq \delta_3 \leq 0.05w_2. \end{aligned} \quad (17)$$

With these binary variables, we have transformed the piecewise function into a linear form. Although this is the approach we discussed in class, Gurobi has the `Model.addGenConstrPWL()` method which is a built-in function that defines a variable as a piecewise-linear function of another variable (Gurobi Optimization, LLC, 2025). This method allows us to approximate our non-linear cost function using a series of linear segments, and therefore, we implement this method in our code instead of adding additional binary variables to the model.

Furthermore, because the maximum expansion allowed is 20% as a result of the piecewise cost function, we revise the upper bound of $x_{f,z} \leq \min(1.2n_{f,z}, 500)$ to $x_{f,z} \leq 0.2n_{f,z}$.

5.1.2 Location Constraints

As stated in the project description, the recommendation from NYS officials mandates a minimum distance of 0.06 miles between any two facilities, whether new or existing, to prevent over concentration.

Our initial approach to incorporating these new constraints was to introduce additional binary decision variables to control the potential new locations. However, this would result in an extremely large number of variables, significantly increasing the model's computational complexity and runtime.

Instead, we keep the same set of decision variables used in Part 1 and compute an upper bound on the total number of new facilities that can be built in each zip code. This estimation involves two main steps: (1) identifying conflicts between any two facilities and (2) determining the maximum feasible number of new facilities.

Identification of Facility Conflicts

The first step in our analysis was to identify all facility conflicts by examining each zip code z classified as a child care desert. The main steps of the process are summarized below:

1. For each zip code $z \in Z$, we define the following parameters:
 - $Exist_z$: the set of existing facilities
 - Pot_z : the set of potential new locations extracted from the initial dataset
 - Pot'_z : the set of all potential locations that are located more than 0.06 miles apart from each existing facility in zip code z
2. Compute the distance $d(i, j)$ for every unique pair of potential locations (i, j) , where $i, j \in Pot'_z$ and $i \neq j$.
3. Define a conflict pair when the distance between two potential locations falls below the minimum threshold of $\delta = 0.06$ miles. Then collect all conflicting pairs in the set C_z for each zip code z , formally defined as:

$$C_z = \{(i, j) \mid i, j \in Pot'_z, i \neq j, \text{ and } d(i, j) < \delta\}.$$

4. The resulting conflict set C_z , which contains all conflicts between two potential locations is then used as input for the next stage of the analysis.

Upper Bound Optimization Model

With these conflict pairs identified, we formulated a preliminary optimization model for each zip code z to find the maximum number of new facilities that could be built.

Decision Variables: For each valid potential location $i \in Pot'_z$, we defined a binary decision variable x_i , where $x_i = 1$ if a new facility is built at location i , and $x_i = 0$ otherwise.

Objective Function: We defined the objective function of this preliminary model to maximize the total number of new facilities selected:

$$\max \sum_{i \in Pot'_z} x_i$$

Constraint: For every pair of potential locations (i, k) that are part of our conflict set (i.e., $d(i, k) < 0.06$ miles), we added the following constraint:

$$x_i + x_k \leq 1$$

This will ensure that at most one of the two conflicting locations can be selected.

Integration into Main Model

Solving this optimization problem for every zip code in our dataset, the resulting optimal objective value, B_z^* , represents the maximum number of new facilities that zip code z can accommodate while respecting the 0.06 mile distance rule.

This value B_z^* is then added as a key constraint in our main formulation for part 2 in the following way:

$$0 \leq B_{s,z} + B_{m,z} + B_{l,z} \leq B_z^* \quad (18)$$

5.2 Formulation

The formulation of this problem uses the same set of decision variables as in Problem 1, which we do not rewrite here to avoid repetition and save space. However, we now define a new cost function, $\text{Cost}'_{E,z}$, using the updated $c_{f,z}$ from Section 5.1.1. For each zip code z , we define

$$\text{Cost}'_{E,z} = \sum_{f \in F_z} c_{f,z}$$

Finally, we include the additional constraint from Section 5.1.2 that limits the number of buildings that can be constructed in each zip code, using the upper bound B_z^* .

With these definitions in place, we can now introduce the complete mathematical formulation of the problem.

$$\begin{aligned} \min \quad & \sum_{z \in Z} (\text{Cost}_{B,z} + \text{Cost}'_{E,z} + \text{Cost}_{5,z}) \\ \text{s.t.} \quad & 0 \leq U_z \leq 50B_{s,z} + 100B_{m,z} + 200B_{l,z} \quad \forall z \in Z, \\ & B_{s,z} + B_{m,z} + B_{l,z} \leq B_z^* \quad \forall z \in Z, \\ & 0 \leq E_{f,z} \leq x_{f,z} \quad \forall f \in F_z \text{ and } \forall z \in Z, \\ & 0 \leq x_{f,z} \leq 0.2n_{f,z} \quad \forall f \in F_z \text{ and } \forall z \in Z, \\ & \text{ExistingSlots}_z + \text{NewSlots}_z \geq \frac{P_z}{2} \quad \forall z \in Z_h, \\ & \text{ExistingSlots}_z + \text{NewSlots}_z \geq \frac{P_z}{3} \quad \forall z \in Z_n, \\ & \text{ExistingSlots}_{5,z} + U_z + \sum_{f \in F} E_{f,z} \geq \frac{2}{3}P_{5,z} \quad \forall z \in Z \\ & B_{s,z}, B_{m,z}, B_{l,z}, E_{f,z}, x_{f,z}, U_z \in \mathbb{Z}^+ \end{aligned} \quad (19)$$

As previously mentioned, the formulation of Problem 2 is very similar to that of Problem 1. The main differences are the introduction of a new cost function for facility expansion and an upper bound, B_z^* , on the number of buildings that can be constructed in each zip code.

In the next section, we present and discuss the results obtained from solving Problem 2.

Please refer to `OptimizationBudgeting_Problem2.ipynb` for the code and `optimization_result_problem2_fin.csv` for the output file of our model.

5.3 Results

For Problem 2, the 20% expansion capacity constraint, a piecewise cost function, and a 0.06-mile minimum distance requirement altogether lead to a different optimization strategy and a higher total funding requirement.

The total funding for this model is **\$268,278,835.82**, representing a 21.2% increase from the total funding in Problem 1. Construction costs remain the majority of the budget but have now increased to 85.2%, while expansion costs decrease significantly from 27.2% of the budget to only 2.6%. Equipment costs make up 12.2% of the budget compared to 14.7% from the first model. The optimized solution requires an 89.1% increase in total new slots for both expansion and construction from 539,207 ($364,700 + 174,507$) in part 1 to now 705,237 ($689,700 + 15,537$). These results show that new construction is strongly preferred, especially for large buildings, in order to effectively eliminate all child care deserts.

Table 3: Results Summary for Problem 2

Metric	Value
Zip Codes Covered	1,313
Desert Elimination Rate	100.0%
Small Facilities (100 slots)	573
Medium Facilities (200 slots)	254
Large Facilities (400 slots)	1,454
Total Slots from Construction	689,700
New U5 Capacity	326,153 slots
Existing Capacity	269,790 slots
Expanded Capacity	15,537 slots

Table 4: Cost Analysis for Problem 2

Cost Category	Amount	Percentage
Construction Cost	\$228,585,000	85.2%
Expansion Cost	\$7,078,535.82	2.6%
Equipment Cost	\$32,615,300	12.2%
Total Cost	\$268,278,835.82	100.0%

6 Comparative Analysis and Case Studies

6.1 Cost Comparison

The cost breakdown and comparison in Table 5 reveal differences between the two optimization approaches for the elimination of the 1,331 childcare deserts in New York.

Table 5: Cost Comparison Analysis between Problem 1 and Problem 2

Metric	Problem 1	Problem 2	Change
Total Cost	\$222,168,837.22	\$268,278,835.82	+20.75%
Construction Cost	\$129,085,000 (58.1%)	\$228,585,000 (85.2%)	+77.08%
Expansion Cost	\$60,468,537 (27.2%)	\$7,078,535.82 (2.6%)	-88.29%
Equipment Cost	\$32,615,300 (14.7%)	\$32,615,300 (12.2%)	0%
Cost per Slot	\$412	\$380	-7.78%

Key Insights

- Problem 2 shows a stronger shift toward construction, which makes up approximately 85% of the total funding cost compared to around 58% in Problem 1. Expansion costs also decrease significantly by approximately 88%, while construction costs increase by a little over 77%. This happens because the updated expansion cost function raises the cost as expansions get larger, pushing the model to substitute toward new builds.
- Total equipment cost remain the same for both models as it pertains to the number of slots specifically for children under the age of 5.
- Even though the total cost increases by 20.75%, the cost-per-slot decreases by -7.78%. This is due to the increase in total new slots as a result of the model's preference for building larger facilities in Problem 2.

6.2 Case Study Analysis on Results

To better understand our results, we looked at four specific zip codes to examine how the overall strategy changes across Problems 1 and 2.

6.2.1 Case Study 1: Expansion-to-Construction Shift (Zip Code 10001)

Metric	Problem 1 Solution	Problem 2 Solution
New Facilities	0 (0 slots)	2 large (800 slots)
New U5 Capacity from Construction	0 slots	400 slots
Existing Center Capacity	609 slots	609 slots
Total Expanded Capacity	496 slots	96 slots
Total Cost	\$119,200	\$358,837.07
<i>Construction</i>	\$0	\$230,000
<i>Expansion</i>	\$156,798.14	\$78,237.07
<i>Equipment</i>	\$49,600	\$49,600

Key Changes: Problem 1 recommends only the expansion of existing facilities with no new facilities to be built. In Problem 2, however, there is a complete shift in strategy from expansion-heavy to construction-heavy. There is a 201% cost increase with construction costs increasing from \$0 in Problem 1 to \$230,000 in Problem 2, while the number of expanded slots decreases from 496 to 96.

6.2.2 Case Study 2: Significant Facility Increase (Zip Code 10457)

Metric	Problem 1 Solution	Problem 2 Solution
New Facilities	3 large (1,200 slots)	21 large (8,400 slots)
New U5 Capacity from Construction	600 slots	4,195 slots
Existing Center Capacity	6,559 slots	6,559 slots
Total Expanded Capacity	3,924 slots	329 slots
Total Cost	\$1,195,621.30	\$3,010,880
<i>Construction</i>	\$345,000	\$2,415,000
<i>Expansion</i>	\$1,168,531.96	\$99,270.58
<i>Equipment</i>	\$452,400	\$452,400

There is a large increase in the number of new facilities to build (600% increase in number of slots) with a corresponding reduction in expansion capacity of existing centers from 3,924 to 329 slots.

6.2.3 Case Study 3: Mixed Strategy (Zip Code 10005)

Metric	Problem 1 Solution	Problem 2 Solution
New Facilities	1 medium + 1 large (600 slots)	2 large (800 slots)
New U5 Capacity from Construction	300 slots	323 slots
Existing Center Capacity	39 slots	39 slots
Total Expanded Capacity	23 slots	0 slots
Total Cost	\$258,694.87	\$262,300
<i>Construction</i>	\$210,000	\$230,000
<i>Expansion</i>	\$16,394.87	\$0
<i>Equipment</i>	\$32,300	\$32,300

There is a shift from a combination of building new medium and large facilities to large facilities only. There is also the elimination of expanding any existing child care center in Problem 2.

6.2.4 Case Study 4: No Change (Zip Code 10524)

Metric	Problem 1 Solution	Problem 2 Solution
New Facilities	1 medium (200 slots)	1 medium (200 slots)
New U5 Capacity from Construction	88 slots	88 slots
Existing Center Capacity	16 slots	16 slots
Total Expanded Capacity	0 slots	0 slots
Total Cost	\$103,800	\$103,800
<i>Construction</i>	\$95,000	\$95,000
<i>Expansion</i>	\$0	\$0
<i>Equipment</i>	\$8,800	\$8,800

Some zip codes like 10524 require minimal changes to existing capacity. Even with additional constraints in Problem 2, the optimal solution for 10524 remains the same for both scenarios.

6.2.5 Analysis Across Cases

1. With the piecewise cost function, large-scale expansions are more expensive, which causes the model to prefer building new facilities over expanding existing ones.
2. The 0.06 mile distance constraint prevents new facility locations from being built in childcare center dense areas. As a result, the model shifts towards expanding existing facilities in these areas.
3. The strategy for some of the zip codes, as seen with Zip code 10524, remains the same across both models. This implies that new construction was already the best choice due to existing infrastructure.

7 Conclusion

This project demonstrates how we can leverage optimization modeling to drive decision-making in the elimination of child care deserts across New York. Through our data cleaning and modeling methods, we developed two models with a 100% elimination rate that addresses two different sets of problems. While expansion and distance constraints to our model increase overall funding required, the resulting solution shows that it is also dependent on the existing conditions and infrastructure of each zipcode. Overall, this work demonstrates how to deliver a more feasible and equitable distribution of child care centers in the state of New York. These findings can help policymakers better understand how to effectively allocate resources to allow for increased access to child care.

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